

Thomas Saurel

Engineering student in Applied Mathematics, Computer Science and Finance at Grenoble INP - ENSIMAG

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I am a second-year student of the engineering cycle at Grenoble INP - ENSIMAG, specializing in Computer Science, Applied Mathematics and Finance. I enjoy working on low-level programming and mathematics with particular interests in quantitative finance, artificial intelligence and computer graphics.

EDUCATION

Engineering Degree <i>Grenoble INP - ENSIMAG</i>	2026-2028
- Track: Fundamentals and Innovative Technologies for Finance - Planned double degree: Master's degree in Quantitative Finance (starting 2027 at Grenoble IAE)	
Double Bachelor's (Mathematics and Computer Science) <i>Université de Toulouse</i>	2023-2026
French General Baccalaureate (Très Bien / European English Section)	2023

SKILLS

Data / ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, XGBoost

Programming Languages: Python, C++23, C, Java, OCaml, SQL; basic knowledge of R and Rocq

Tools: Git, Linux, LaTeX

UNIVERSITY PROJECTS

Fourier Transforms for Image Compression	2025-2026
- Toulouse Institute of Mathematics (supervisor: <i>Philippe Monnier</i>) - Study of Fourier transforms, image spectra and the Cooley-Tukey FFT algorithm - Python implementation of FFT and DCT algorithms for JPEG-inspired image compression	
Motion Tracking for Piano Playing	2025-2026
- Hand motion detection during piano playing using MMPose and MediaPipe - Object detection, trajectory analysis and trajectory prediction	
ENEDIS Data Challenge	2025
- Development of algorithms for missing data imputation in time series - Data completion using regression models (XGBoost) and neural networks (DAE, LSTM) - Achieved a 24% reduction in MAE compared with the linear interpolation benchmark	
Arc Length of Parametric and Regular Curves	2024-2025
- Toulouse Institute of Mathematics (supervisor: <i>Patrice Lassère</i>) - Theoretical study of the length of parametrized and regular curves, ellipses and geodesics	

PERSONAL PROJECTS

Study and Implementation of the Cox-Ross-Rubinstein Model	Ongoing
- Study of no-arbitrage, replication and risk-neutral pricing - Pricing of European and American options using backward induction - Design and implementation in C++23	
Ray Tracer in C++20	2026
- Implemented the full Ray Tracing in One Weekend series (P. Shirley, 3 books): BVH, textures, volumetrics, importance sampling (Monte Carlo) - Modernized the code to C++20, parallelized rendering with std::thread and reduced pointer-related overhead	
Video Game Development	2018-2024
- Data structures and decision-making AI algorithms (State Machines, Behavior Trees) - Object-oriented programming (Java, GDScript, UE4) and 3D modeling with Blender	

LANGUAGES

French : Native

English : B2 - Cambridge English Certificate (2023) - TOEFL planned in December

Spanish : B2